

A CO-REQ APPENDIX: INTEGER POWERS OF 10

Nonnegative Integer Powers of 10

The phrase **nonnegative integers** refers to the set containing 0, 1, 2, 3, ... and so on. In the expression 10^5 , 10 is called the **base**, and 5 is called the **exponent**, or **power**. The exponent 5 is telling us to multiply the base 10 by itself 5 times. So, $10^5 = 10 \times 10 \times 10 \times 10 \times 10 = 100,000$. By definition, any number raised to the 0 power is 1. So, $10^0 = 1$.

In the following table, there are several nonnegative integer powers of 10 that have been written as a product. Notice that higher exponents result in larger products. What do you notice about the number of zeros in the resulting product?

Exponential Form	Product	Number of Zeros in Product
10^0	1	0
10^1	10	1
10^2	$10 \times 10 = 100$	2
10^3	$10 \times 10 \times 10 = 1,000$	3
10^4	$10 \times 10 \times 10 \times 10 = 10,000$	4
10^5	$10 \times 10 \times 10 \times 10 \times 10 = 100,000$	5

That's right! The number of zeros is the same as the power each time!

Negative Integer Powers of 10

The **reciprocal** of a number is 1 divided by that number. For example, the reciprocal of 10 is $\frac{1}{10}$. We use negative exponents to indicate a reciprocal. For example, $10^{-1} = \frac{1}{10^1} = \frac{1}{10}$. Similarly, any expression with a negative exponent can be written with a positive exponent by taking the reciprocal. Several negative powers of 10 have been simplified in the table that follows. What do you notice about the number of zeros in the denominator (bottom) of each fraction?

Exponential Form	Equivalent Simplified Expression	Number of Zeros in Denominator
10^{-1}	$\frac{1}{10^1} = \frac{1}{10}$	1
10^{-2}	$\frac{1}{10^2} = \frac{1}{10 \times 10} = \frac{1}{100}$	2
10^{-3}	$\frac{1}{10^3} = \frac{1}{10 \times 10 \times 10} = \frac{1}{1,000}$	3
10^{-4}	$\frac{1}{10^4} = \frac{1}{10 \times 10 \times 10 \times 10} = \frac{1}{10,000}$	4

That's right! The number of zeros is the same as the positive version of the power each time.

In the following table, we will write the same powers of 10 as decimals. Count the number of decimal places to the right of the decimal point. What do you notice?

Exponential Form	Equivalent Simplified Expression	Number of Decimal Places to Right of Decimal
10^{-1}	$\frac{1}{10^1} = 1 \div 10 = 0.1$	1
10^{-2}	$\frac{1}{10^2} = 1 \div 100 = 0.01$	2
10^{-3}	$\frac{1}{10^3} = 1 \div 1,000 = 0.001$	3
10^{-4}	$\frac{1}{10^4} = 1 \div 10,000 = 0.0001$	4

That's right! The number of decimal places to the right of the decimal point is the same as the positive version of the power each time.

Multiplying Integers by Positive Powers of 10

Did you know that the distance from the sun to Earth is over 90 million miles? This value can be represented as 90,000,000, or we can write it as a product: $9 \times 10,000,000 = 9 \times 10^7$, which is actually a more compact way of writing 90 million. Notice that the power of 7 reflects the number of zeros in 90 million. Several products of positive integers and powers of 10 are given in the table that follows. Notice that the number of zeros is the same as the exponent except in one case.

Exponential Form	Product	Number of Zeros in Product
5×10^1	$5 \times 10 = 50$	1
13×10^2	$13 \times 100 = 1,300$	2
8×10^3	$8 \times 1,000 = 8,000$	3
15×10^4	$15 \times 10,000 = 150,000$	4
70×10^5	$70 \times 100,000 = 7,000,000$	6

The only case in which the number of zeros didn't equal the exponent was the last case. Why do you think that happened? That's right! We multiplied by 70 which also had a zero. So, the product had a zero from the 70 and 5 zeros from 10^5 for a total of 6 zeros in 7,000,000.

Multiplying by Negative Powers of 10

As we have seen, negative powers of 10 are decimals. Several products of positive integers and powers of 10 are given in the table below. Notice that multiplying an integer by 10 raised to a negative integer power results in a smaller number than you started with. Also, the number of decimal places to the right of the decimal point is the same as the exponent except in one case.

Exponential Form	Product	Number of Decimal Places to Right of Decimal
3×10^{-1}	$3 \times 0.1 = 0.3$	1
13×10^{-2}	$13 \times 0.01 = 0.13$	2
9×10^{-3}	$9 \times 0.001 = 0.009$	3

Exponential Form	Product	Number of Decimal Places to Right of Decimal
15×10^{-4}	$15 \times 0.0001 = 0.0015$	4
70×10^{-5}	$70 \times 0.00001 = 0.00070$ or 0.0007	5(6 if we leave on the extra 0)

The only case in which the number of decimal places to the right of the decimal point didn't equal the positive version of the exponent was the last case. Why do you think that happened? That's right! We multiplied by 70, which ended in zero.

Moving the Decimal Place

A helpful shortcut when multiplying a number by a power of 10 is to “move the decimal point.” The following table shows several powers of 10, both positive and negative. Compare the location of the decimal point in the original number to the location of the decimal point in the product. How has it changed?

Exponential Form	Product	How the Position of the Decimal Point Changed
5×10^1	$5. \times 10 = 5 \underset{\wedge}{} 0. = 50$	1 place to the right
13×10^2	$13. \times 100 = 13 \underset{\wedge}{} 0 = 1,300$	2 places to the right
8×10^3	$8. \times 1,000 = 8 \underset{\wedge}{} 0 = 8,000$	3 places to the right
15×10^4	$15. \times 10000 = 15 \underset{\wedge}{} 0 = 150,000$	4 places to the right
70×10^5	$70. \times 100,000 = 70 \underset{\wedge}{} 0 = 7,000,000$	5 places to the right
3×10^{-1}	$3. \times 0.1 = . \underset{\wedge}{} 3 = 0.3$	1 place to the left
13×10^{-2}	$13. \times 0.01 = . \underset{\wedge}{} 1 3 = 0.13$	2 places to the left
9×10^{-3}	$9. \times 0.001 = . \underset{\wedge}{} 0 9 = 0.009$	3 places to the left
15×10^{-4}	$15. \times 0.0001 = . \underset{\wedge}{} 0 1 5 = 0.0015$	4 places to the left
70×10^{-5}	$70. \times 0.00001 = . \underset{\wedge}{} 0 7 0 = 0.0007$	5 places to the left

Notice that multiplying by a positive power of 10 moves the decimal point to the right, making the value larger, while multiplying by a negative power of 10 moves the decimal point to the left, making the value smaller. Also, the number of decimal places that the decimal point moves is exactly the positive version of the exponent.

